

GROMOS Peptide Workflow: File Roles and Dependency Chain

Purpose

This document summarizes the GROMOS peptide workflow as a dependency tree. It lists which files are inputs, scripts, and outputs; what each file does; and how the files depend on one another. The emphasis is on three physically important stages:

- adding missing hydrogens and completing the structure,
- stabilization by energy minimization and equilibration,
- neutralizing the system with chloride ions so the total charge is zero.

Folder Structure

The workflow is organized into the following folders:

<code>top/</code>	topology construction and final topology files
<code>mix_top/</code>	topology building blocks, checks, and combined topology
<code>coord/</code>	coordinate preparation and hydrogen completion
<code>min/</code>	vacuum minimization
<code>box/</code>	solvation
<code>Solvent_min/</code>	restrained minimization of solvated system
<code>ion/</code>	ionization / charge neutralization
<code>eq/</code>	equilibration / thermalization
<code>viewing/</code>	visualization and movie generation

File Roles by Stage

1. Topology generation

- **Inputs:** 54a7.mtb, 54a7.ifp, make_top_peptide.arg, make_top_Cl.arg, com_top_peptide_2Cl.arg, check_top_peptide.arg
- **Commands:**
 - make_top @f make_top_peptide.arg
 - make_top @f make_top_Cl.arg
 - com_top @f com_top_peptide_2Cl.arg
 - check_top @topo peptide.top
- **Outputs:** peptide.top, CL_54a7.top, peptide_2Cl_54a7.top

This stage defines the system chemistry. The final combined topology peptide_2Cl_54a7.top is the topology used once chloride ions are part of the system.

2. Coordinate preparation

- **Inputs:** peptide.pdb, pdb2g96_peptide.arg, ghc_peptide.arg
- **Commands:**
 - pdb2g96 @f pdb2g96_peptide.arg
 - gch @f ghc_peptide.arg
- **Outputs:** pdb2g96_peptide.cnf (or peptide.cnf), peptide_fixed_h.cnf

This stage has the most important structural correction role:

- **pdb2g96** converts the PDB file into GROMOS coordinate format.
- **gch** completes the coordinates by adding missing hydrogens and fixing atoms that were incomplete or placed at the origin.

This is the **hydrogen completion** stage.

3. Vacuum minimization

- **Inputs:** peptide_fixed_h.cnf, em_peptide.imd, em_peptide.run, peptide_2Cl_54a7.top
- **Command:** ./em_peptide.run
- **Outputs:** peptide_min.cnf, em_peptide.umd

This stage relaxes the peptide in vacuum and removes bad contacts. It is the first stabilization step, and it produces a physically more reasonable starting structure.

This is the **geometry stabilization** stage.

4. Solvation

- **Inputs:** peptide_min.cnf, sim_box_peptide.arg, spc.cnf, peptide_2Cl_54a7.top
- **Command:** sim_box @f sim_box_peptide.arg
- **Output:** sim_box_peptide.cnf

This step inserts the minimized peptide into a box of SPC water.

5. Solvent minimization with restraints

- **Inputs:** sim_box_peptide.cnf, sim_box_peptide.por, sim_box_peptide.rpr, em_solvent.imd, em_solvent.run, peptide_2Cl_54a7.top
- **Command:** ./em_solvent.run
- **Outputs:** peptide_h2o.cnf, em_solvent.umd

The .por file contains the atoms to restrain (solute only). The .rpr file contains the reference positions. This stage relaxes the water around the peptide while keeping the peptide restrained.

This is the **solvent stabilization** stage.

6. Ionization

- **Inputs:** `ion_peptide.arg`, `peptide_h2o.cnf`, `peptide_2Cl_54a7.top`
- **Command:** `ion @f ion_peptide.arg`
- **Output:** `peptide_2Cl_h2o.cnf`

This stage replaces water molecules with two chloride ions. The purpose is to neutralize the total charge of the system.

This is the **charge neutralization** stage.

7. Equilibration

- **Inputs:** `peptide_2Cl_h2o.cnf`, `peptide_2Cl_h2o.por`, `peptide_2Cl_h2o.rpr`, `equilibration.imd`, `equilibration.jobs`, `eq_mk_script.arg`, `mk_script.lib`, `peptide_2Cl_54a7.top`
- **Commands:** `mk_script @f eq_mk_script.arg`, then `./eq_peptide_1.run`
- **Outputs:** `eq_peptide_1.run` to `eq_peptide_5.run`, `eq_peptide_*.imd`, `eq_peptide_*.cnf`, `eq_peptide_*.trc.gz`, `eq_peptide_*.tre.gz`, `eq_peptide_*.omd`

This is the thermal stabilization stage. The system is heated gradually from low temperature to 300 K while restraints are loosened step by step.

8. Visualization

- **Inputs:** `eq_peptide_*.trc.gz` or `eq_peptide_*.trc`, `peptide_2Cl_54a7.top`, `view_peptide.arg`
- **Commands:** `frameout`, VMD rendering, `ffmpeg`
- **Outputs:** `FRAME_00001.pdb`, rendered images such as `simulation.00001.ppm`, final movie `movie.mp4`

Detailed Dependency Chain

The chain below shows the logic in the order it is executed:

1. `peptide.pdb`

2. `pdb2g96_peptide.cnf` generated by `pdb2g96`
3. `peptide_fixed_h.cnf` generated by `gch` (**hydrogen completion**)
4. `peptide_min.cnf` generated by vacuum minimization (**geometry stabilization**)
5. `sim_box_peptide.cnf` generated by `sim_box`
6. `peptide_h2o.cnf` generated by restrained solvent minimization (**solvent stabilization**)
7. `peptide_2Cl_h2o.cnf` generated by `ion` (**charge neutralization**)
8. `eq_peptide_1.run` to `eq_peptide_5.run` generated by `mk_script`
9. `eq_peptide_5.cnf` and `eq_peptide_5.trc.gz` generated by equilibration (**thermal stabilization**)
10. `FRAME_00001.pdb` generated by `frameout`
11. rendered image sequence such as `simulation.00001.ppm`
12. final movie `movie.mp4` generated with `ffmpeg`

What each key stage accomplishes

Hydrogen completion

`gch` adds the missing hydrogens and repairs incomplete coordinates.

Geometry stabilization

vacuum minimization removes bad contacts and relaxes the peptide structure.

Solvent stabilization

restrained minimization lets the water relax around the peptide.

Charge neutralization

ionization adds two chloride ions so the system has zero net charge.

Thermal stabilization

equilibration heats the system gradually and removes restraints step-wise.

Minimal replication checklist

To reproduce the workflow later, keep the following folder order and files:

- `top/`: build topology files
- `coord/`: PDB conversion and hydrogen completion
- `min/`: vacuum minimization
- `box/`: solvation
- `Solvent_min/`: restrained solvent minimization
- `ion/`: ionization to neutralize charge
- `eq/`: equilibration and thermal stabilization
- `viewing/`: frame extraction and movie creation